

MAT1002 Lecture 7,

Outline

- Parametric plane curves
- Tangents / lengths / areas
- Polar coordinates

Parametric Plane Curves

In MAT1001, we studied the computation of quantities, such as lengths and areas, associated to a curve if the curve is described as the graph of a function $y = f(x)$. In this chapter we look at curves that have a more general description.

Definition

If a plane curve has the form

$$\{(x, y) : x = f(t), y = g(t), t \in I\},$$

where I is an interval, then the curve is called a **parametric curve**, and the equations

$$x = f(t), y = g(t), \textcircled{t} \in I$$

t is called the parameter

are called **parametric equations** of the curve.

e.g. (a) If a curve is the graph of a function $y = f(x)$, where $x \in I$, then it is (trivially) a parametric curve, with parametric equations

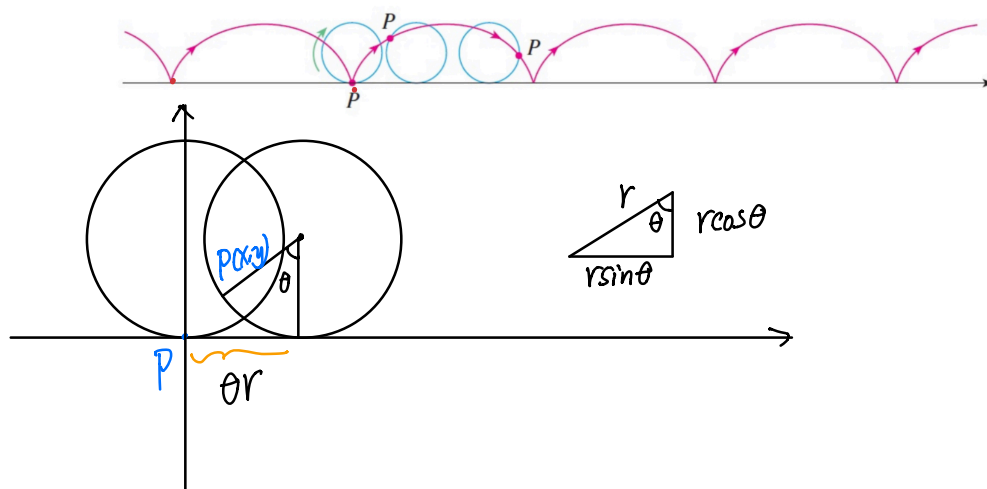
$$\underline{x = t}, \underline{y = f(t)}, t \in I.$$

e.g. (b) Sketch the curve

$$x = \sin t, y = \sin^2 t, t \in \mathbb{R}$$

Note that the same point set of a curve can be given by different parametric equations. Given two different sets of parametric equations to the curve above.

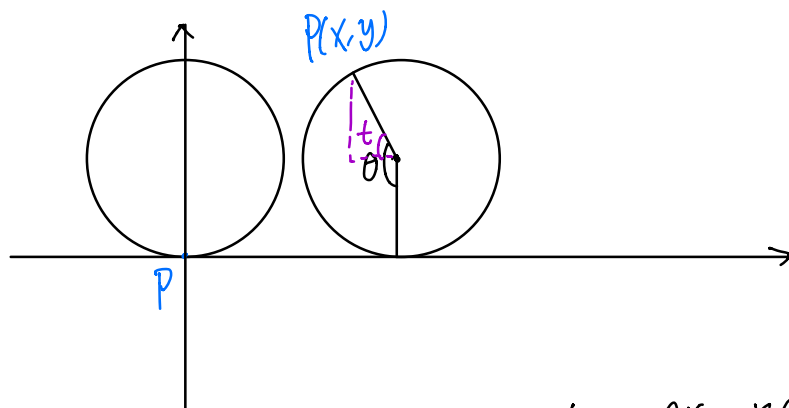
e.g. (c) The curve traced out by a point P on a disk as the disk rolls along a straight line is called a **cycloid** (see the figure below). Assume that the disk has radius r and rolls along the x -axis. If one position of P is the origin, find parametric equations for the cycloid.



For $\theta \in [0, \frac{\pi}{2}]$:

$$x = \theta r - r \sin \theta = r(\theta - \sin \theta)$$

$$y = r - r \cos \theta = r(1 - \cos \theta)$$



For $\theta \in [\frac{\pi}{2}, \pi]$:

$$x = \theta r - r \cos t = \theta r - r \cos(\theta - \frac{\pi}{2})$$

$$= \theta r - r \sin \theta = r(\theta - \sin \theta)$$

$$y = r + r \sin t = r + r \sin(\theta - \frac{\pi}{2})$$

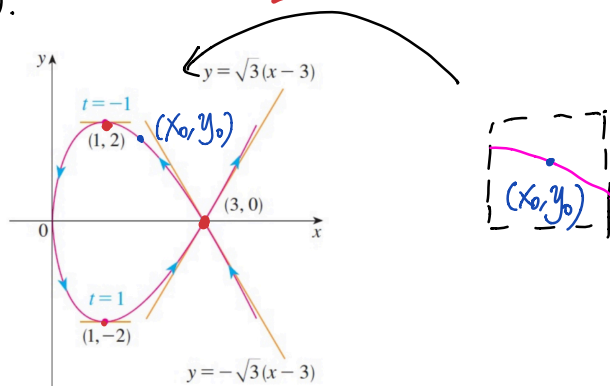
$$= r + r(-\cos \theta) = r(1 - \cos \theta)$$

Using similar arguments for $\theta \in [\pi, \frac{3\pi}{2}]$ and $\theta \in [\frac{3\pi}{2}, 2\pi]$, one can show that the cycloid can be parametrized by

$$x = r(\theta - \sin\theta), \quad y = r(1 - \cos\theta), \quad \theta \in \mathbb{R}.$$

Tangents

Consider the curve given by $x = t^2$, $y = t^3 - 3t$, $t \in \mathbb{R}$, which is sketched below.



How can one find the slope of the tangent at the point (x_0, y_0) ? Note that near the point, the curve is the graph of $y = h(x)$. More generally, if $x = f(t)$, $y = g(t)$, and y is a differentiable function of x (near a point), then

$$\underline{\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dx}{dt}} \Rightarrow \frac{dy}{dx} = \frac{dy/dt}{dx/dt} \quad \left(\text{if } \frac{dx}{dt} \neq 0 \right).$$

expressed in terms of t

For the example above, $\frac{dy}{dx} \Big|_{(x_0, y_0)}$ = $\frac{3t^2 - 3}{2t} \Big|_{t=t_0}$ where $(x_0, y_0) = (f(t_0), g(t_0))$.

Second Derivatives

Similarly, if $x = f(t)$, $y = g(t)$, and y is a twice differentiable function of x near a point, then at the point,

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{d}{dt} x} = \frac{\frac{d}{dt} \left(\frac{dy/dt}{dx/dt} \right)}{\frac{dx}{dt}}$$

For the example above, at $(x_0, y_0) = (f(t_0), g(t_0))$.

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{\frac{d}{dt} \left(\frac{3t^2 - 3}{2t} \right)}{\frac{dx}{dt}} = \frac{\frac{d}{dt} \left(\frac{3}{2} \left(t - \frac{1}{t} \right) \right)}{2t} = \frac{\frac{3}{2} \left(1 + \frac{1}{t^2} \right)}{2t} \\ &= \frac{3(t^2 + 1)}{4t^3} \end{aligned}$$

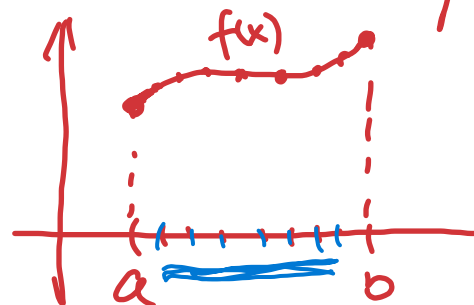
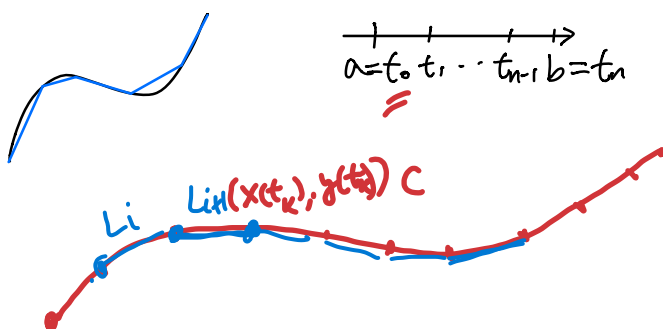
Curve is
Concave up

< 0, if $t < 0$ ← Concave down

Arc Lengths

Suppose C is given by $x = f(t)$, $y = g(t)$, $t \in [a, b]$. Assume that f' and g' are continuous, and C is traversed exactly once. We may approximate the length of C , as follows.

$$L = \int_a^b \sqrt{1 + f'^2} dx$$



$$(x_2, y_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\Delta x_i = x_i - x_{i-1} = f(t_i) - f(t_{i-1})$$

$$\Delta y_i = y_i - y_{i-1} = g(t_i) - g(t_{i-1})$$

$$L_i = \sqrt{\Delta x_i^2 + \Delta y_i^2} = \sqrt{(f'(c_i) \Delta t_i)^2 + (g'(d_i) \Delta t_i)^2}$$

$$\Delta x_i = \underset{MVT}{f'(c_i) \Delta t_i}, \text{ where } c_i \in (t_{i-1}, t_i)$$

Similarly, $\Delta y_i = g'(d_i) \Delta t_i, d_i \in (t_{i-1}, t_i)$.

$$\text{Length of } C: L \approx \sum_{i=1}^n L_i = \sum_{i=1}^n \sqrt{f'(c_i)^2 + g'(d_i)^2} \Delta t_i$$

It can be shown formally using upper and lower sums that (if f' and g' are cts)

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt{f'(c_i)^2 + g'(d_i)^2} \Delta t_i = \int_a^b \sqrt{f'(t)^2 + g'(t)^2} dt$$

This motivates the following definition $= \int_a^b \sqrt{x'(t)^2 + y'(t)^2} dt$

Definition

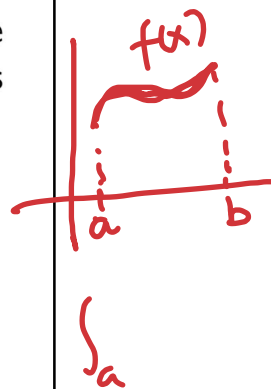
Let C be a curve given by $x = f(t), y = g(t)$ and $t \in [a, b]$, where f' and g' are continuous on $[a, b]$. If C is traversed exactly once as t increases from a to b (except that $t = a$ and $t = b$ may give the same point), then the **length** L of C is defined by

$$L := \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \int_a^b \sqrt{(f'(t))^2 + (g'(t))^2} dt$$

$$= \int_a^b \sqrt{1 + f'^2} dx$$

$$y = f(x)$$

$$\begin{cases} x = t \\ y = f(t) \end{cases} \begin{matrix} \frac{dx}{dt} = 1 \\ \frac{dy}{dt} = f' \end{matrix}$$



Example

Find the length of the parametric curve given by

$$x = \sin 2t, \quad y = \cos 2t, \quad 0 \leq t \leq 2\pi.$$

$$\begin{aligned} 0 &\leq t \leq 2\pi \\ 0 &\leq 2t \leq 4\pi \\ x^2 + y^2 &= 1 \end{aligned}$$

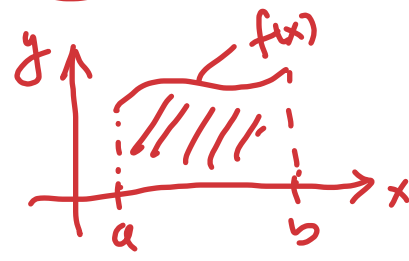


Sol: The curve is the unit circle, but traced twice.

It is traced exactly once when t goes from 0 to π . So

$$L = \int_0^\pi \sqrt{f'(t)^2 + g'(t)^2} dt = \int_0^\pi \sqrt{4\cos^2(2t) + 4\sin^2(2t)} dt$$

$$= 2 \int_0^\pi dt = 2\pi.$$



Areas under Curves

We know that the area under the graph of a nonnegative function $y = h(x)$ from $x = \alpha$ to $x = \beta$ is given by $A = \int_\alpha^\beta h(x) dx$. If the graph is traced out exactly once by the parametric equations

$$x = f(t), \quad y = g(t), \quad a \leq t \leq b,$$

where f' and g' are continuous, then by substitution, we have

$$A = \int_a^b g(t) f'(t) dt \quad \left(\text{or } A = \int_b^a g(t) f'(t) dt \right).$$

if $x \uparrow$ as $t \uparrow$

$$\begin{cases} x=b \\ y=g(t)=g(x) \end{cases}$$

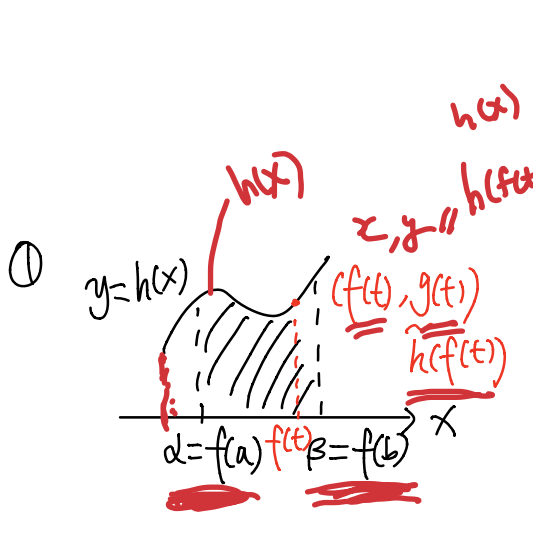
$$f'(t)=1$$

if $x \downarrow$ as $t \uparrow$

$$\int_a^b g(x) dx$$

$$a \leq x \leq b$$

$$\int_a^b f(x) dx$$

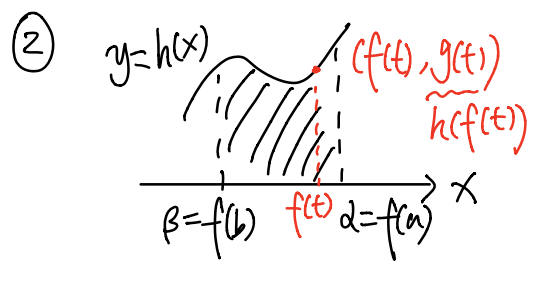
①  $y=h(x)$ $h(x)$ $h(f(t))$ $(f(t), g(t))$ $\tilde{h}(f(t))$ x $a=f(a)$ $b=f(b)$

$g(t)=h(f(t))$ on the curve.

$$A = \int_a^b h(x) dx \quad x=f(t) \quad dx=f'(t)dt$$

$$= \int_a^b h(f(t)) f'(t) dt$$

$$= \int_a^b g(t) f'(t) dt.$$

②  $y=h(x)$ $(f(t), g(t))$ $\tilde{h}(f(t))$ x $b=f(b)$ $a=f(a)$

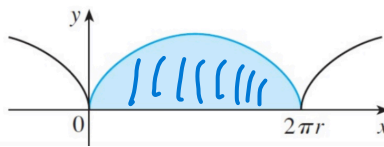
$\forall x \rightarrow$ as $t \uparrow$, then

$$A = \int_b^a h(x) dx$$

$$= \int_b^a g(t) f'(t) dt.$$

Example

Find the area under one arch of the cycloid.



$$\begin{cases} x=r(\theta - \sin\theta) \\ y=r(1 - \cos\theta) \end{cases}$$

$\theta \in [0, 2\pi]$ $x'=r(1 - \cos\theta)$

$$A = \int_0^{2\pi} \underbrace{r(1 - \cos\theta)}_{g(\theta)} \underbrace{r(1 - \cos\theta)}_{f'(\theta)} d\theta = \underline{\text{exercise}} = \underline{3\pi r^2}.$$

Arc Length Differentials

Let $S(t) := \int_a^t \sqrt{(f'(u))^2 + (g'(u))^2} du$, $\begin{matrix} (f(a), g(a)) \\ (f(t), g(t)) \\ (f(b_1), g(b_1)) \end{matrix}$ $\begin{matrix} (f(b_2), g(b_2)) \\ C: \begin{cases} x=f(t) \\ y=g(t) \end{cases} \\ b_1 \leq t \leq b_2. \end{matrix}$

can be negative

the signed distance from the base point $(f(a), g(a))$ to a point $(f(t), g(t))$ on the curve C .

$T = (f(t), g(t))$

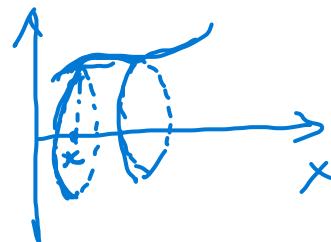
$A = (x(a), y(a)) = (f(a), g(a))$

$$\frac{ds(t)}{dt} = \sqrt{f'(t)^2 + g'(t)^2}$$

By FTC, $\frac{ds}{dt} = \sqrt{f'(t)^2 + g'(t)^2} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$

The differential ds is called the arc length differential,
and

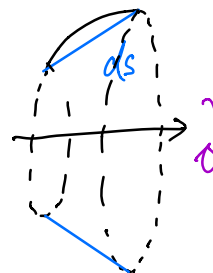
$$ds = \sqrt{f'(t)^2 + g'(t)^2} dt$$



Surfaces of Revolution

In MAT101, we discussed surfaces generated by revolving the graph $y = h(x)$ (≥ 0) ^{about the x-axis} for $\alpha \leq x \leq \beta$. Its area is

$$S = \int_{\alpha}^{\beta} 2\pi y \sqrt{1 + h'(x)^2} dx$$

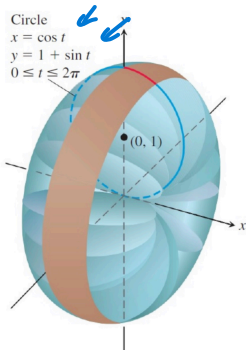


More generally, if the surface is generated by revolving a curve given by $x = f(t)$, $y = g(t)$ (≥ 0) for about the x-axis

$a \leq t \leq b$, its area is

$$S = \int_a^b 2\pi g(t) \sqrt{f'(t)^2 + g'(t)^2} dt$$

Using the formula, the surface area drawn in the following figure can be computed with a straightforward integral. (This is Example 9 in Chapter 11.2 of the book.)



$$S = \int_0^{2\pi} \sqrt{1 + \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$= \int_0^{2\pi} \sqrt{1 + (-\sin t)^2 + (\cos t)^2} dt$$

$$= \int_0^{2\pi} \sqrt{1 + \sin^2 t + \cos^2 t} dt$$

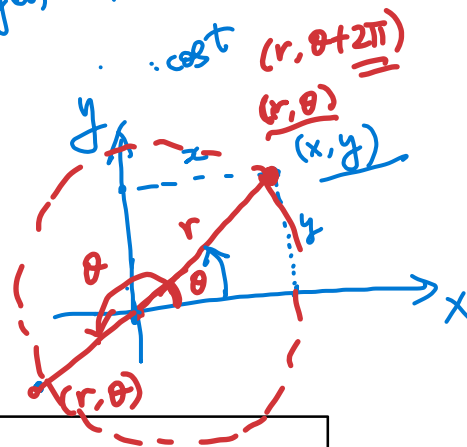
$$= \int_0^{2\pi} \sqrt{2} dt$$

$$= \sqrt{2} \cdot 2\pi = 2\sqrt{2}\pi$$

Exercise.

$$x = f(t) = \cos t$$

$$y = g(t) = 1 + \sin t$$



Polar Coordinates

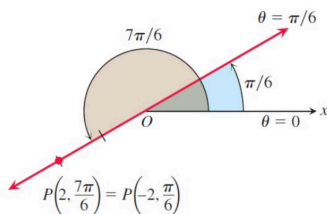
Definition

For a point $(x, y) \in \mathbb{R}^2$ on the xy -plane in Cartesian coordinates:

- ▶ let r be the length of the line segment L joining the points $(0, 0)$ and (x, y) , and;
- ▶ let θ be the angle made by L and the positive x -axis, with the positive sign indicating the counterclockwise measurement.

Then the point (r, θ) is called a polar coordinate of the point (x, y) .

We may extend the definition by allowing "negative distance": if a point has polar coordinate (r, θ) where $r < 0$, we can also represent it by $(-r, \theta + \pi)$, e.g.



(x, y) : Cartesian coordinate, unique.

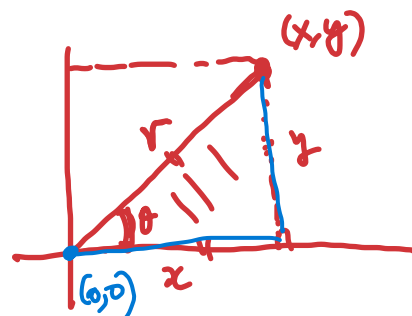
(r, θ) : Polar coordinate, not unique.

$\hookrightarrow$ e.g. $(1, \frac{\pi}{4}) = (1, \frac{9\pi}{4}) = \cancel{(-1, \frac{5\pi}{4})}$
 $= \frac{\pi}{4} + 2\pi = \frac{9\pi}{4}$

Conversion between Cartesian and Polar Coordinates

Given (r, θ) : (x, y) is given by

$x = r \cos \theta$ $y = r \sin \theta$



Given (x, y) : (r, θ) is given by

$r^2 = x^2 + y^2$, $\tan \theta = \frac{y}{x}$ (if $x \neq 0$).

If $x=0$, then $\theta = \frac{\pi}{2}$ if $y \geq 0$, and $\theta = \frac{3\pi}{2}$ if $y < 0$.

(Note that the origin can be given by any angle.)

$\cos(\theta + 2\pi) = \cos \theta$

$\sin(\theta + 2\pi) = \sin \theta$

e.g. Express the curve $r = \frac{4}{2 \cos \theta - \sin \theta}$ with a Cartesian equation.

all points whose polar coordinate satisfy the equation.

$(0, 0) \quad r=0$
 $\theta = \theta_0$

$x^2 + y^2 = a^2$
 $r^2 = a^2$

$r = a$

Sol: $(*) \Leftrightarrow \frac{2r \cos \theta - r \sin \theta}{2x - y} = \frac{4}{4} \Leftrightarrow$
 $\Leftrightarrow \underline{2x - y = 4} \Leftrightarrow$

$y = 2x - 4$

Ans: $y = 2x - 4$

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

e.g. Express the Curve $x^2 + xy + y^2 = 1$ ^(*) with a polar equation.

Sol.: (*) $\Leftrightarrow$ $r^2 + r^2 \cos \theta \sin \theta = 1$

$\Leftrightarrow$ $r^2 (1 + \cos \theta \sin \theta) = 1$

$\Leftrightarrow$ $r^2 = \frac{1}{1 + \cos \theta \sin \theta}$ Ans.